Four equivalent formulations of being a monomorphism in a category

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:mono' (source lines 124-134, Lemma 10.1.1); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The four internal item labels item:mono1, item:mono4, item:mono2 and item:mono3 are defined inside the retained statement and used inside the retained proof, so they need no external entry.

Editorial note. The retained counter values (chapter 10, section 1, theorem offset 0) make the counted lemma print as Lemma 10.1.1, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial scope. One original result of the HoTT Book, counted once: four equivalent formulations of being a monomorphism (Lemma 10.1.1).

Allowed standard prerequisites (editorial).

- Category. A category is a precategory whose hom-types are sets.. It is the hypothesis under which the four conditions are compared, and the set-ness of the hom-types is what makes the fibers propositions.
- Fiber of a function between sets. For a function between sets, injectivity is equivalent to all fibers being mere propositions.. It is the equivalence of conditions (1) and (3) at source line 138.
- Inhabited mere propositions are contractible. A mere proposition that is inhabited is contractible.. It is the step from condition (3) to condition (4) at source line 139.

Lemma 10.1.1. *For a morphism $f : \hom_A(a, b)$ in a category A, the following are equivalent.*

(i) *f is a **monomorphism**: for all $x : A$ and $g, h : \hom_A(x, a)$, if $f \circ g = f \circ h$ then $g = h$.*

(ii) *(If A has pullbacks) the diagonal map $a \to a \times_b a$ is an isomorphism.*

(iii) *For all $x : A$ and $k : \hom_A(x, b)$, the type $\sum_{(h:\hom_A(x,a))}(k = f \circ h)$ is a mere proposition.*

(iv) For all $x : A$ and $g : \hom_A(x, a)$, the type $\sum_{(h:\hom_A(x,a))}(f \circ g = f \circ h)$ is contractible.

Proof. The equivalence of conditions (i) and (ii) is standard category theory. Now consider the function $(f \circ -) : \hom_A(x, a) \to \hom_A(x, b)$ between sets. Condition (i) says that it is injective, while (iii) says that its fibers are mere propositions; hence they are equivalent. And (iii) implies (iv) by taking $k :\equiv f \circ g$ and recalling that an inhabited mere proposition is contractible. Finally, (iv) implies (i) since if $p : f \circ g = f \circ h$, then (g, refl) and (h, p) both inhabit the type in (iv), hence are equal and so $g = h$. □